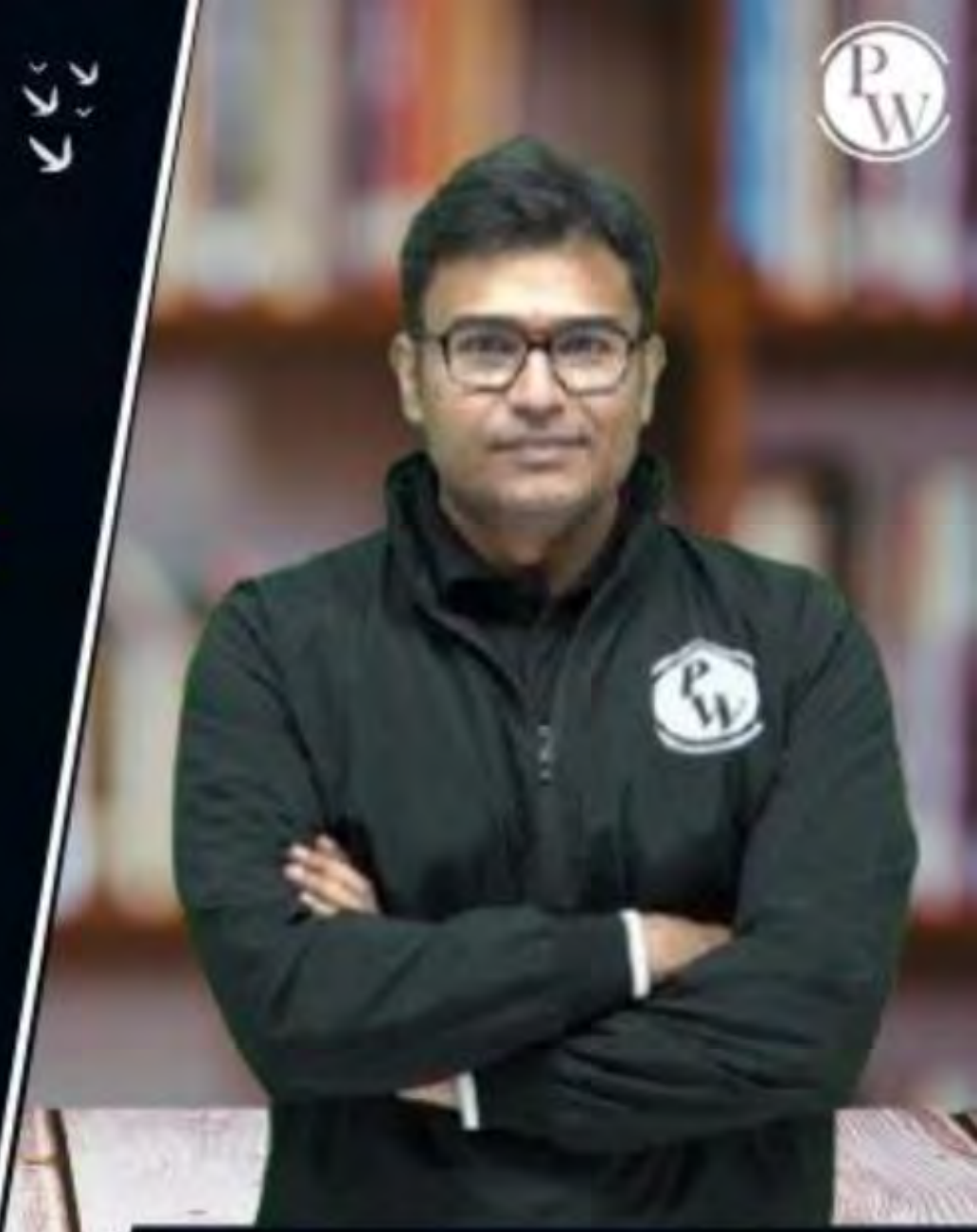
Computer Science & IT

Theory of Computation



CFL

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

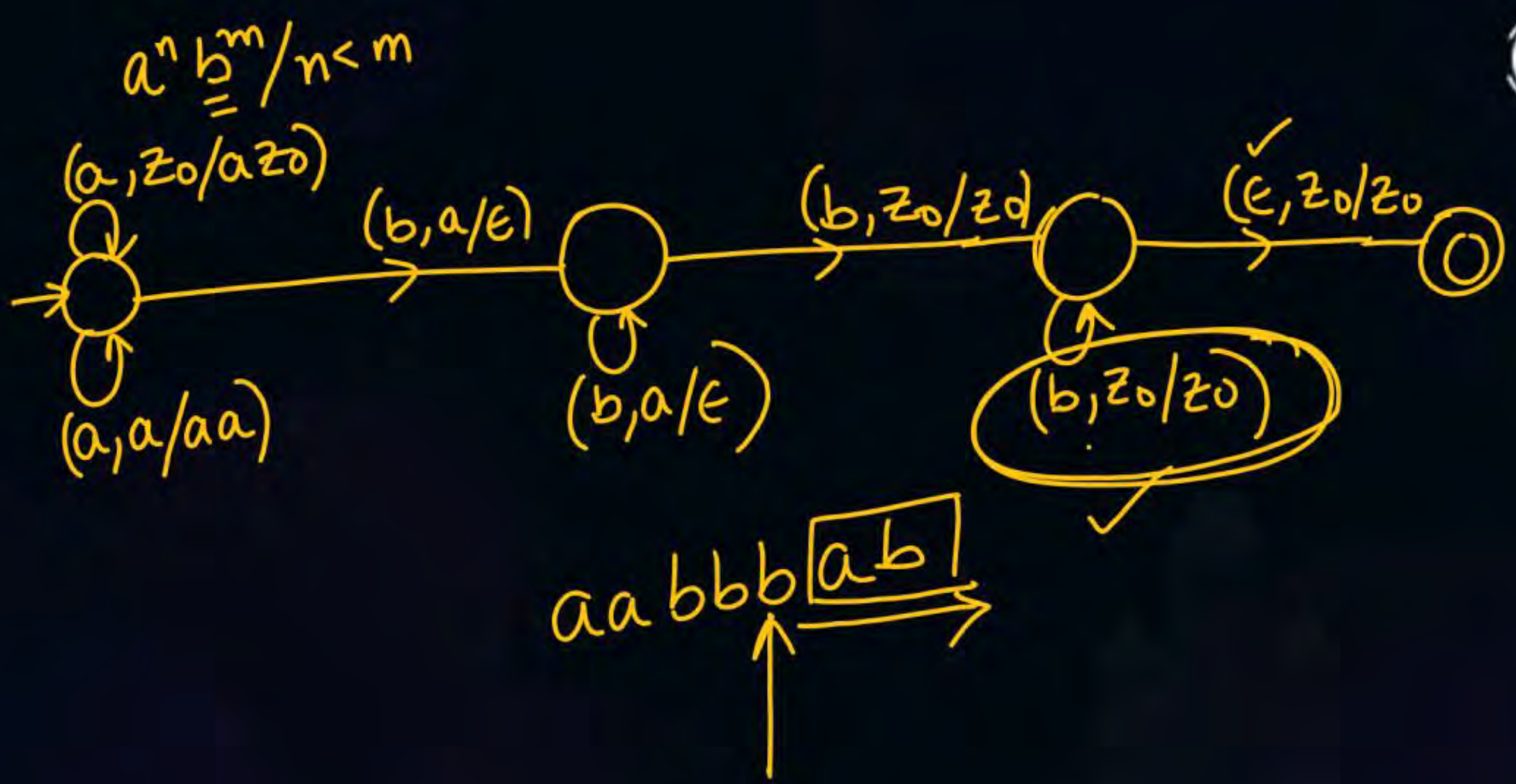
PDA

Topics to be Covered



- Topic
- Topic
- Topic

PDA

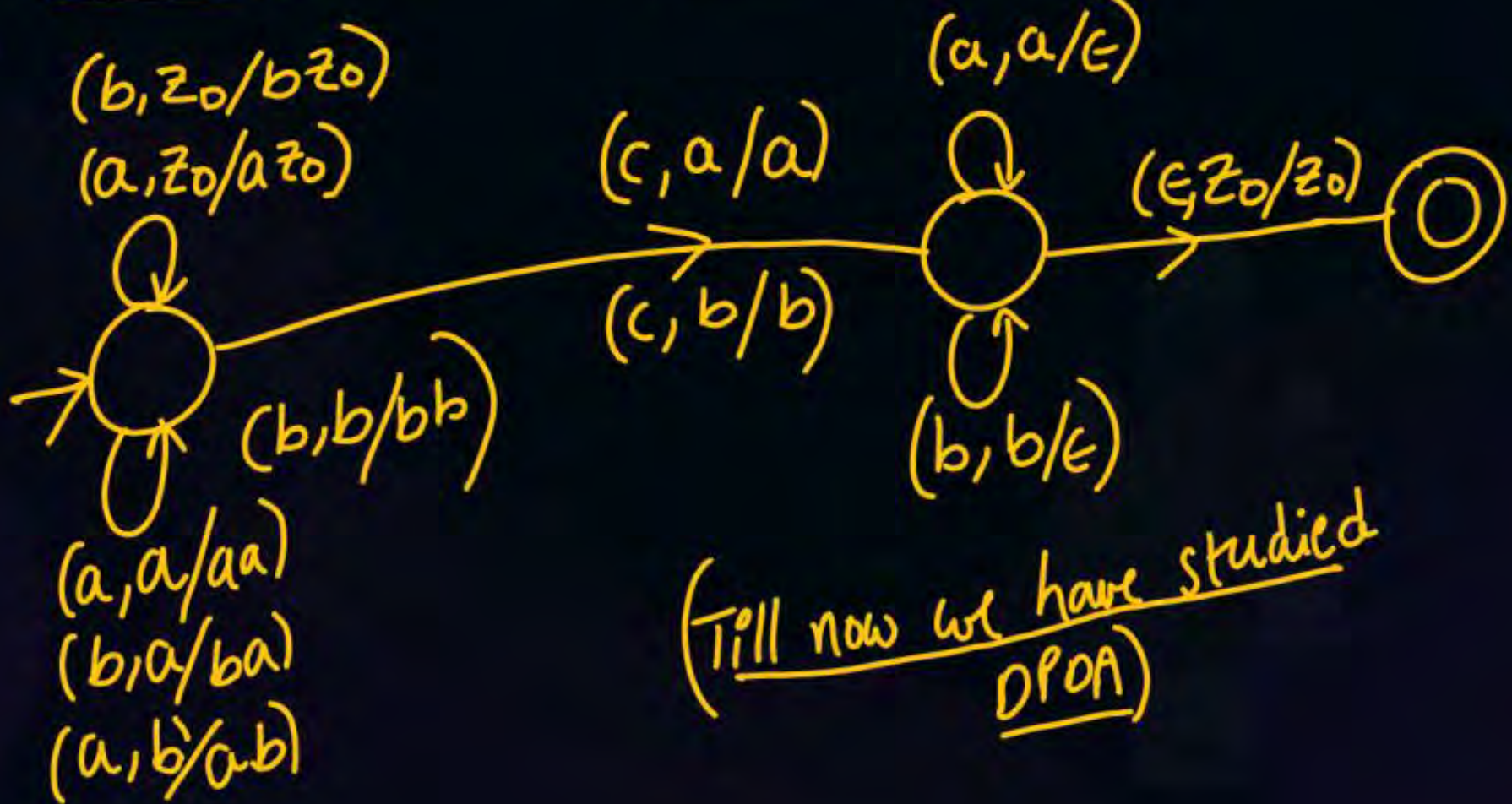

$a^n b^m / n < m$
$n \leq m$
$(a, z_0/az_0)$
$(b, a/\epsilon)$
$(b, z_0/z_0)$
$(\epsilon, z_0/z_0)$
$(a, a/aa)$
$(b, a/\epsilon)$
$(b, z_0/z_0)$
aabbb
ab

→ $\underline{w}c\underline{w}^R / w \in (a+b)^+$, c is a special symbol





Deterministic Pushdown automata



(Till now we have studied DPDA)

w/ $n_a(w) = n_b(w)$

a a b b



a b a b



b b a a



b a b a



abba

Chance → matching → Pop (a, a) (b, a)

NO chance → push (a, a) (b, b).

(a, Z0) (b, Z0)

a b b a

a b b ∈

(b, Z0/b Z0)

(a, Z0/a Z0)

(∈, Z0/Z0)

(a, b/∈)

(b, a/∈)

(a, a/aa)

(b, b/bb)

(∈, b,

min PDA ✗

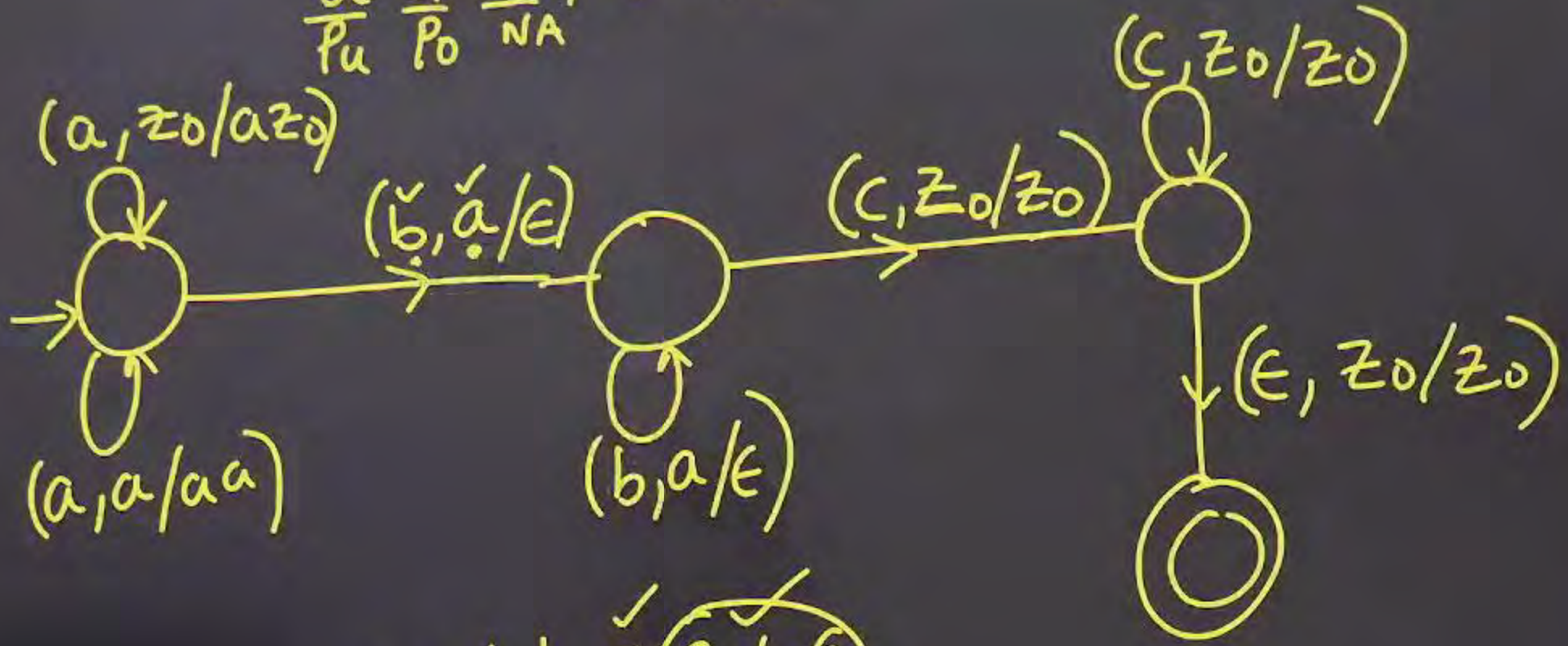
min of PDA ✗

$a^n b^n c^m / n, m \geq 1$

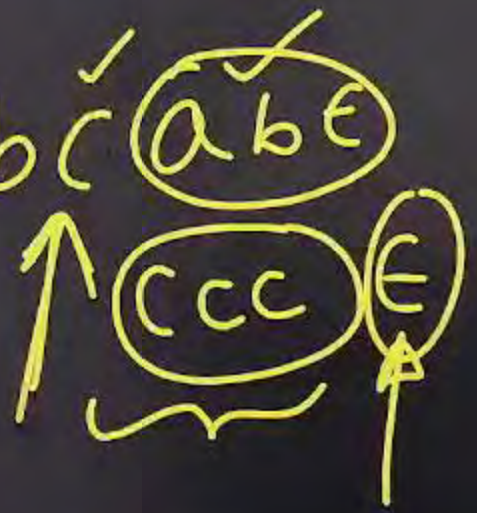
PDA
- acceptance by Empty stack
- acceptance by final stack

$a^n b^n c^m / n, m \geq 1$

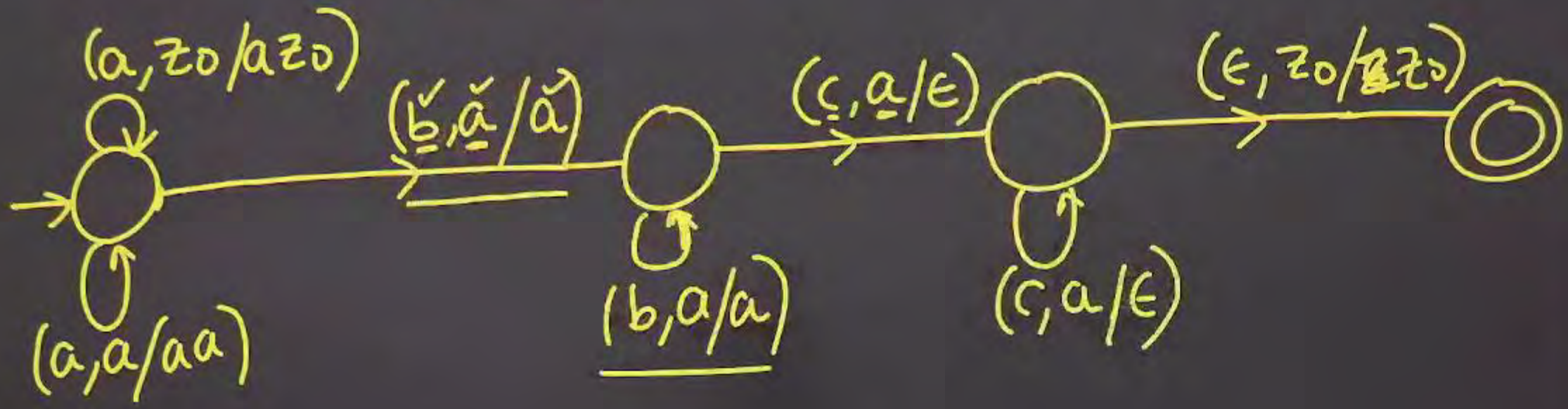
Pu Po NA



aabbčć



$a^n b^m c^n / n, m \geq 1$



$a^n b^n c^n / n \geq 1$

aa bb cc

$a^m b^l c^k$ / $2m = l + k$

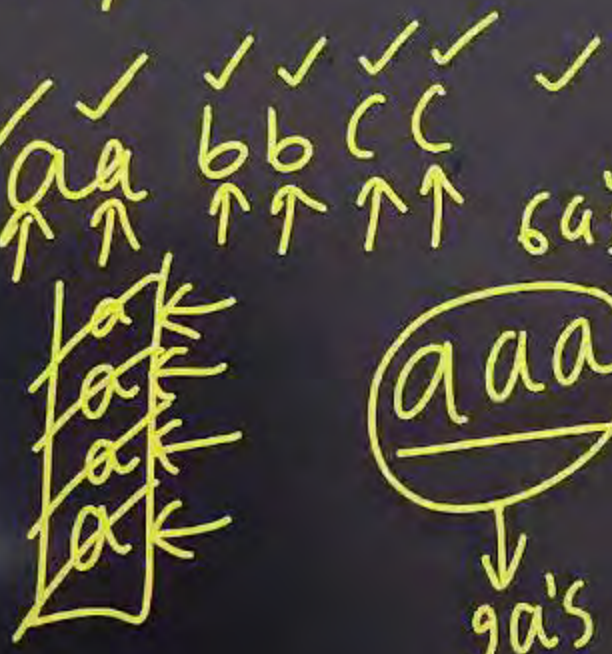
acc

aabbbc

6a's

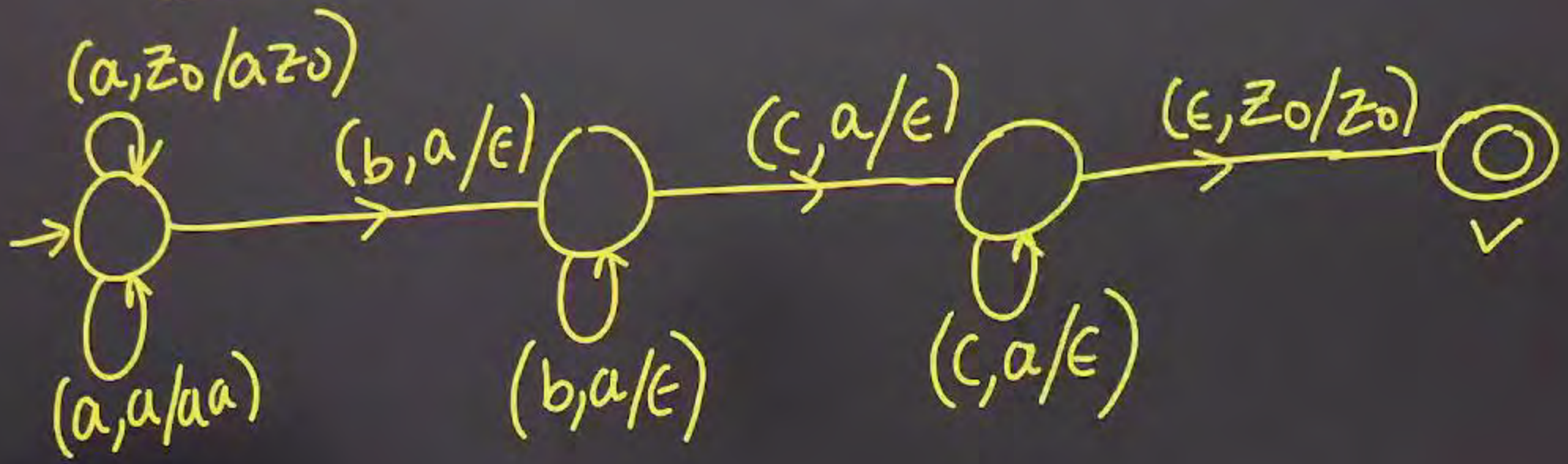
aaa bbcccc

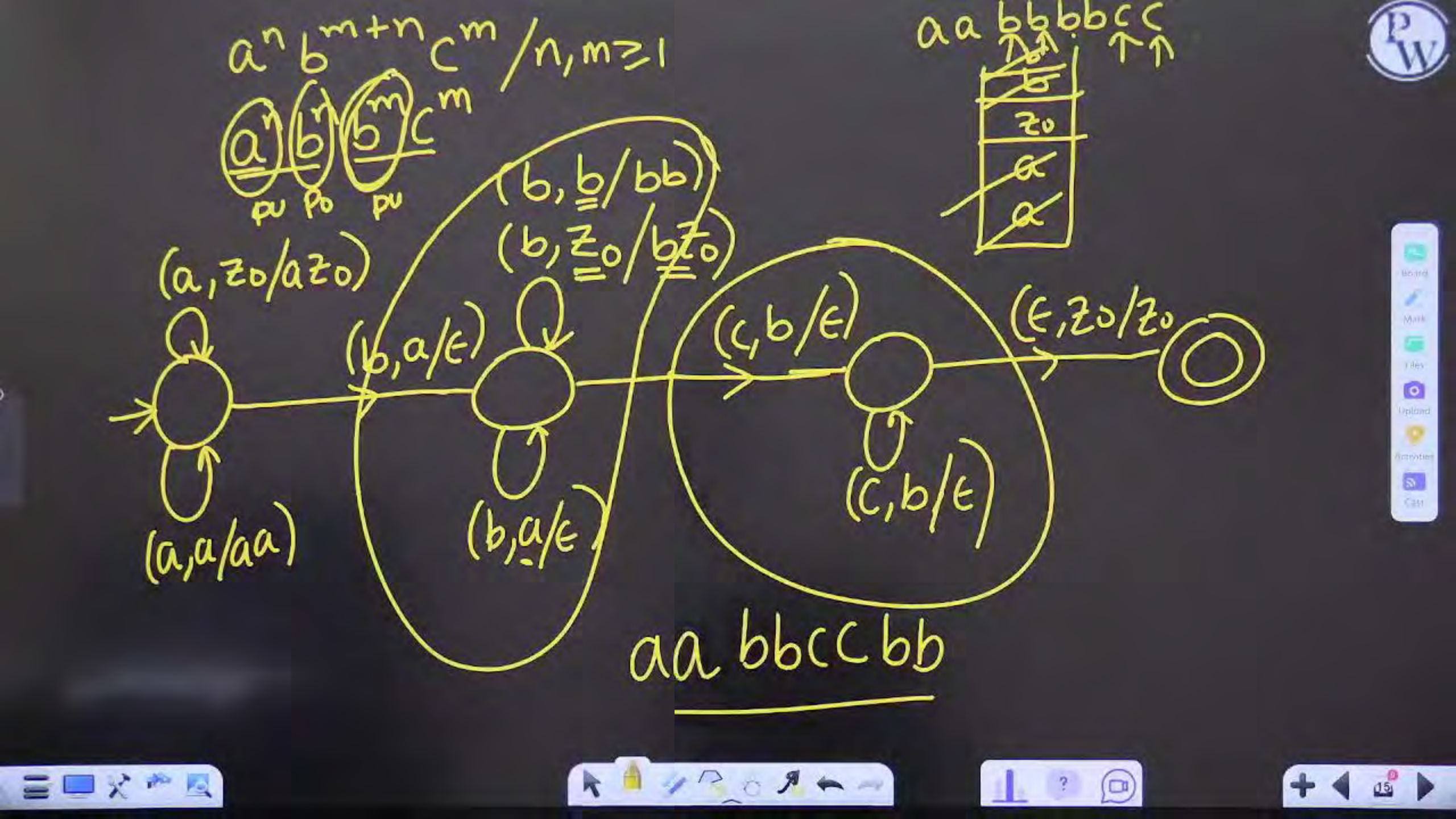
2a's



$a^{n+m} b^m c^n / m,n \geq 1$

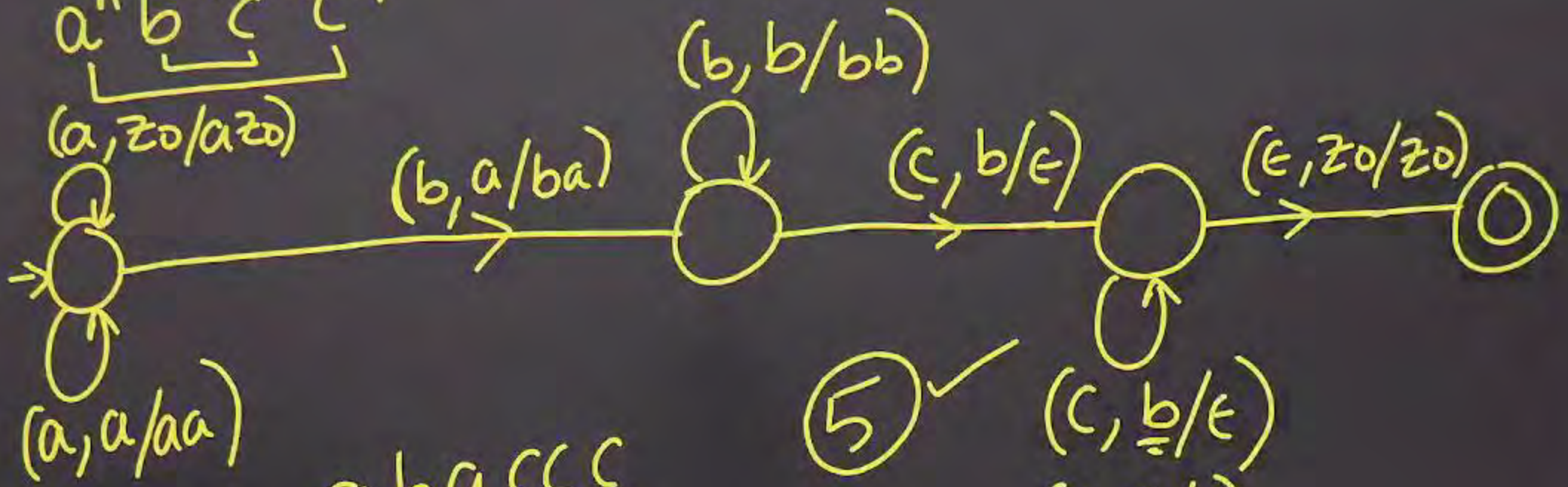
$a^n a^m b^m c^n$




$a^n b^{m+n} c^m / n, m \geq 1$
a^n b^m b^n c^m
PU PO PU
aa bb bb cc
b
Z_0
a
a
(a, Z_0/aZ_0)
(a, a/aa)
(b, a/ϵ)
(b, b/bb)
(b, Z_0/bZ_0)
(b, a/ϵ)
(c, b/ϵ)
(c, b/ϵ)
(ϵ, Z_0/Z_0)
aa bbcc bb

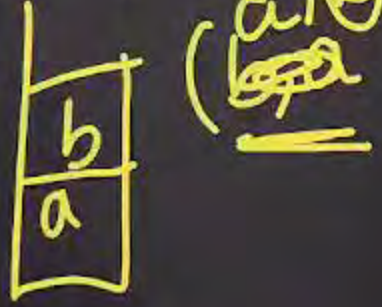
$a^n b^m c^{n+m} / n,m \geq 1$

$a^n b^m c^m c^n$

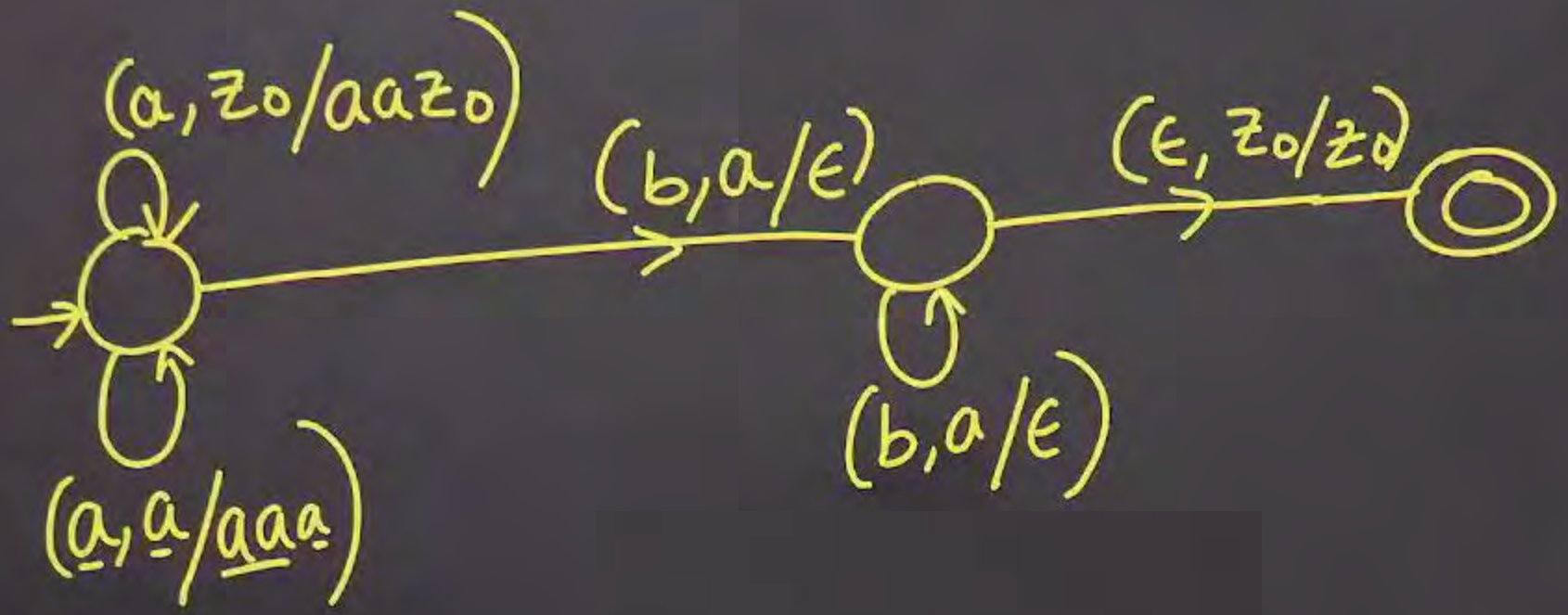


5 ✓

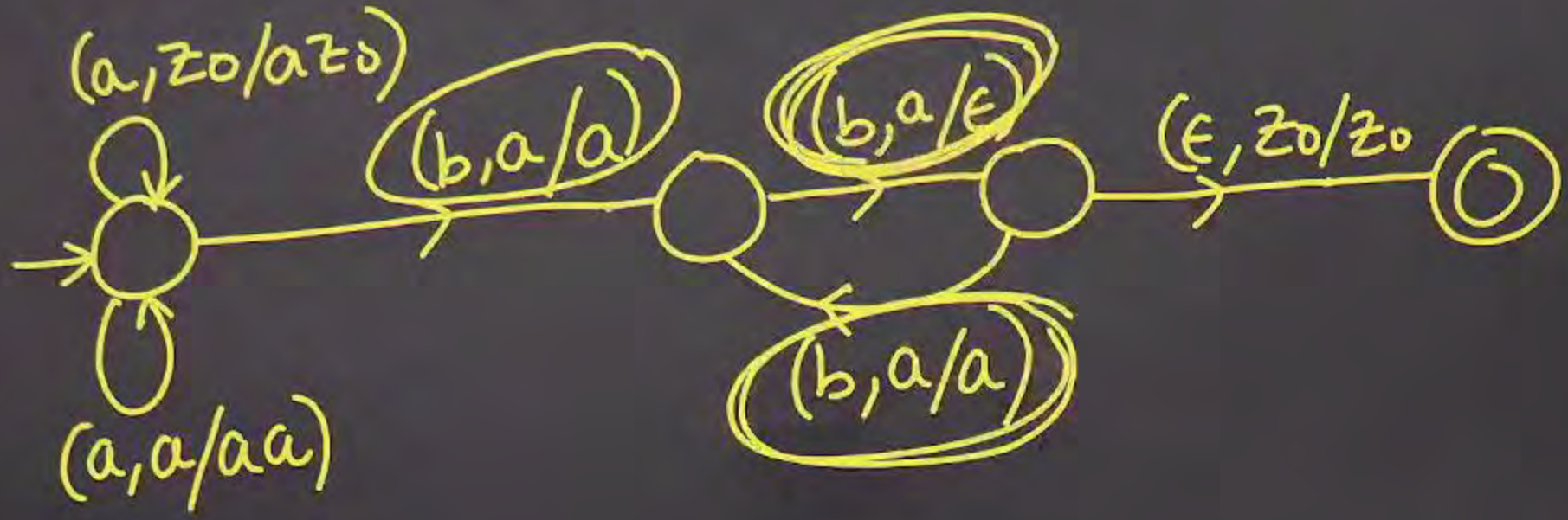
abaccc



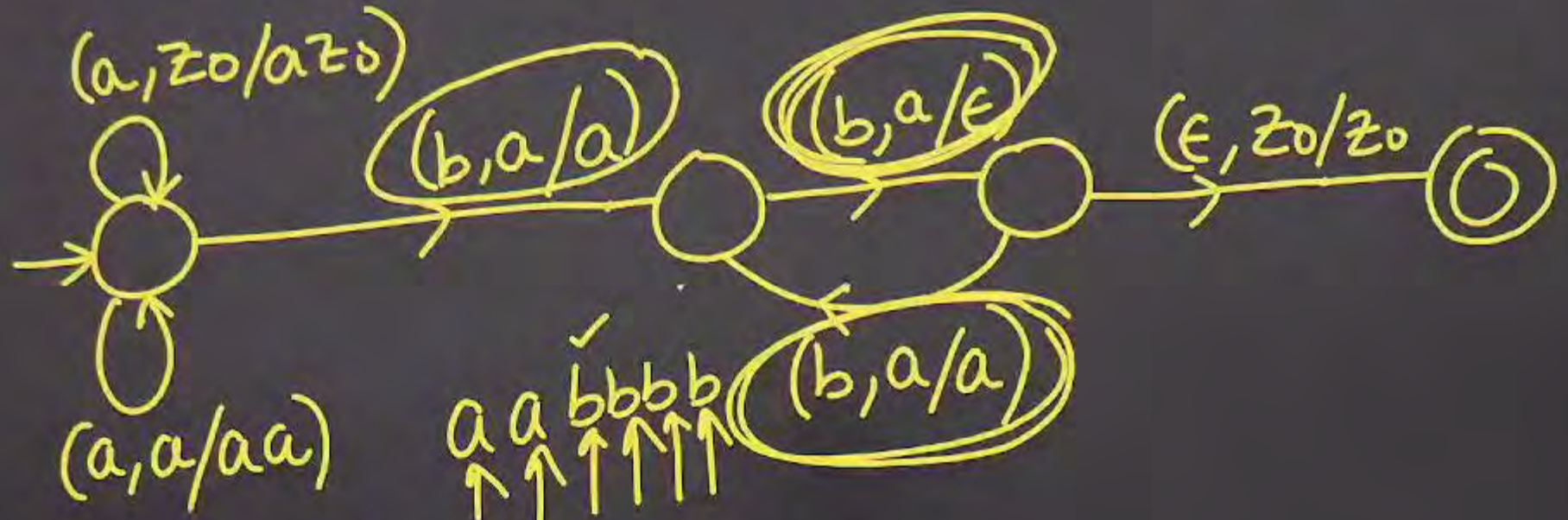
$a^n b^{2n} / n \geq 1$



a → push | b → skip | b → pop a



a → push | b → skip | b → pop a

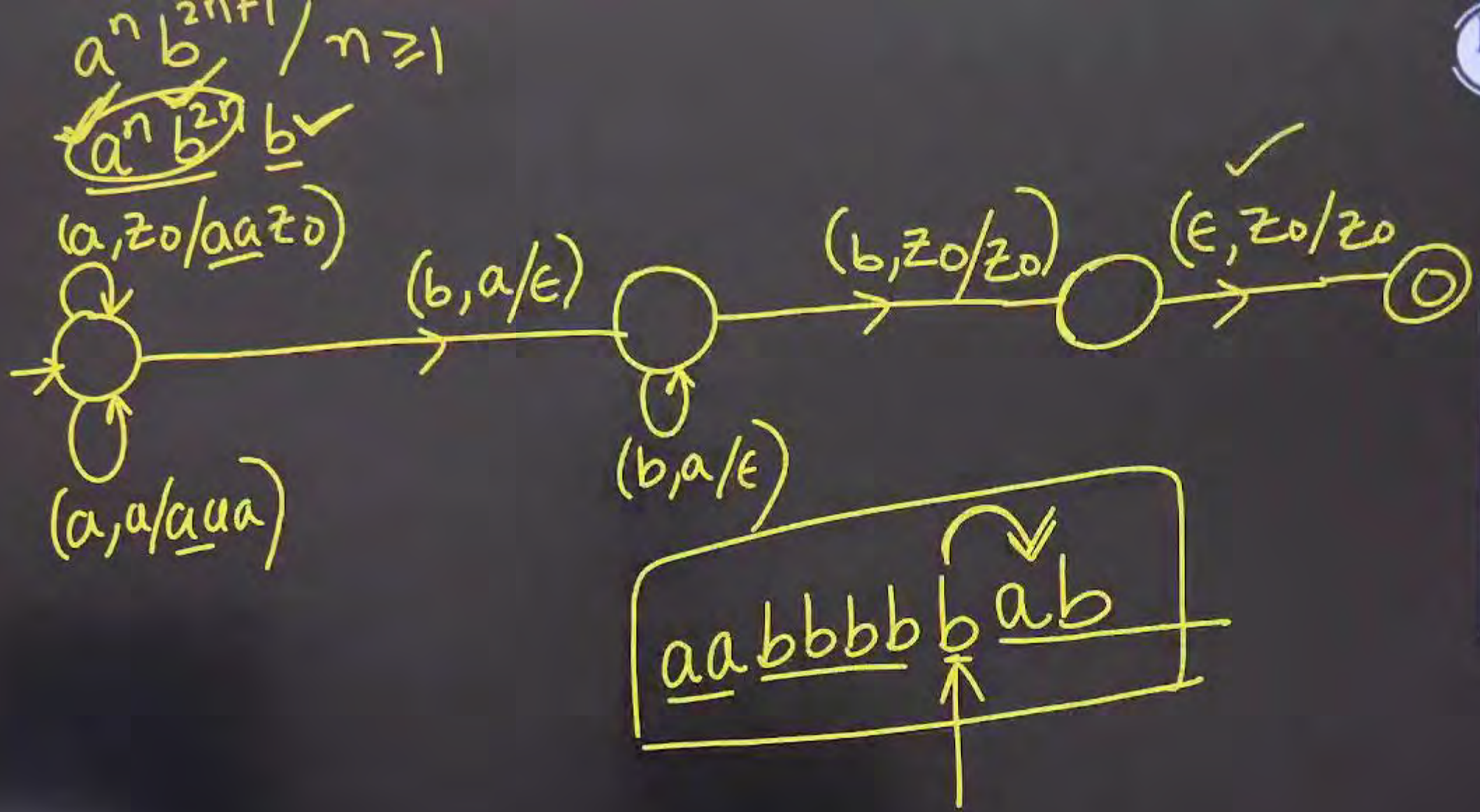


a a b b b b



$a^n b^{2n+1} \quad n \geq 1$

$a^n b^{2n}$ b ✓



aabbbbbab

THANK - YOU